ASL Character Identification

Milestone 3

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Summary of Milestone 2 Progress

RQ: Can a vision model recognize static ASL hand signs from a single RGB image and generalize to signers it has never seen?

- 36 classes: digits 0–9 + letters A–Z

American Sign Language Dataset (Rai, 2020)

- 10 signers × 36 classes × 100 imgs = 36,000 images
- 300×300 RGB JPG
- 1.24 GB on disk (raw + processed variants)

Summary of Milestone 2 Progress

We opt to use the raw images – variety of positions, angles, lighting conditions across images

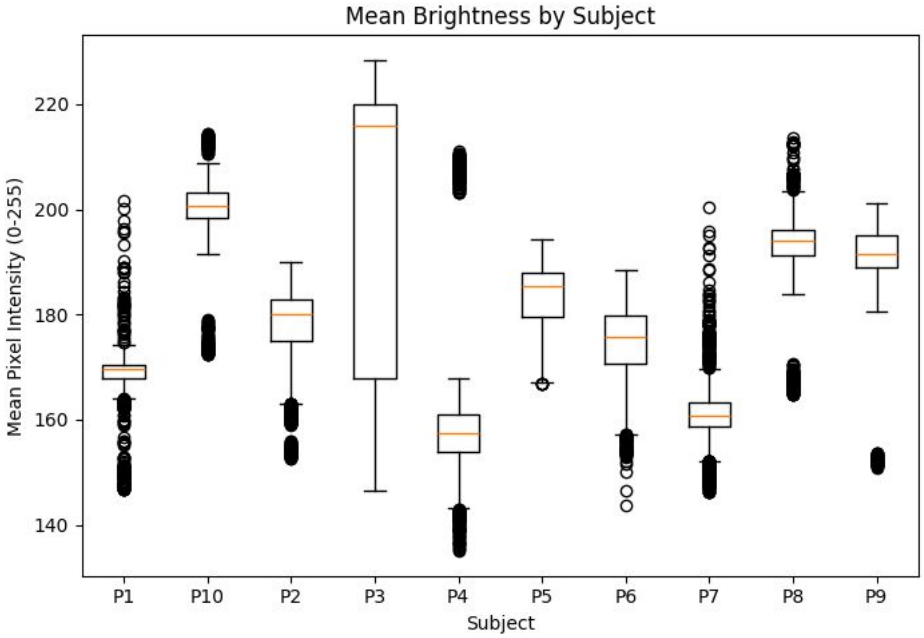
Created our own dataset split by subject: Subjects 1-6 for train, 7 & 8 for validation, 9 & 10 for testing

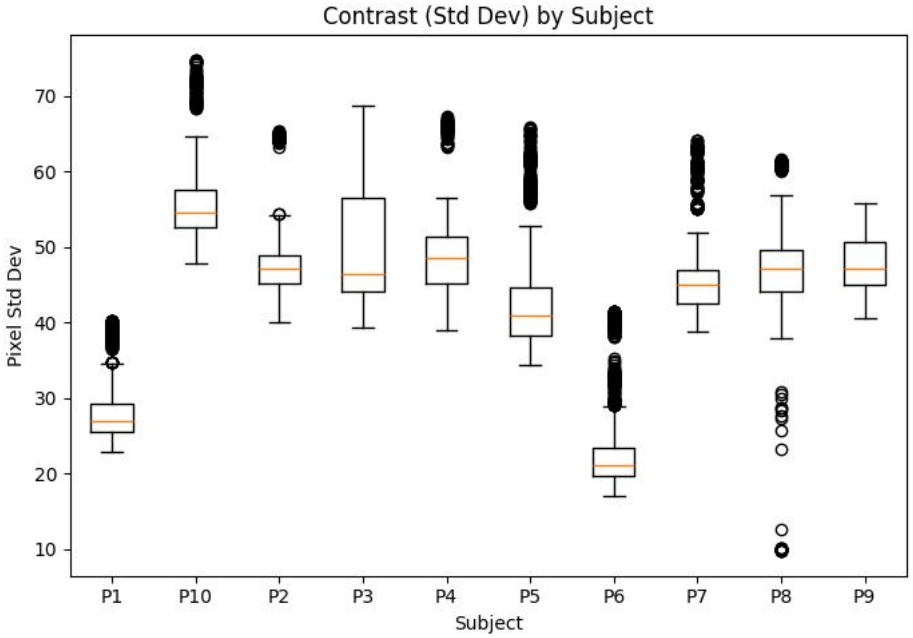
Splitting creates out-of-sample subjects, but we intend to create/find out-of-sample test data

Preprocessing steps:

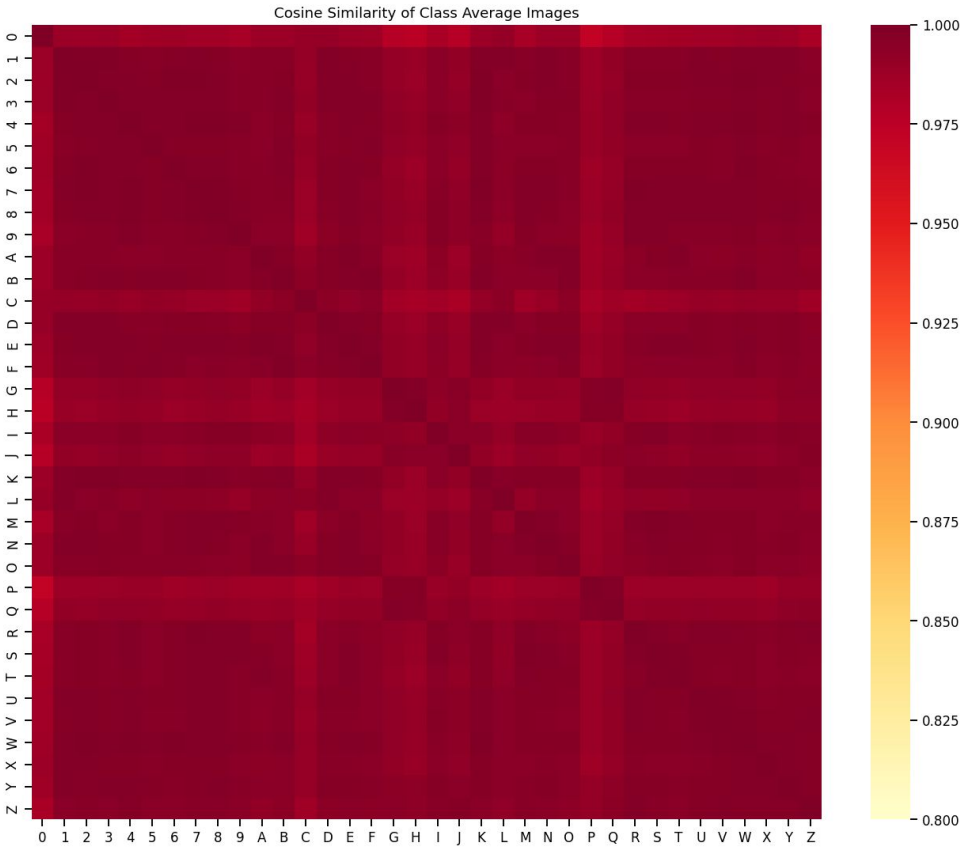
- Random rotation (+- 10°)
- Color jitter
- Normalization for ResNet pretraining

Statistics Per Subject





Class Similarity & Background



Baseline Model 1: Logistic Regression

Input: 32×32 grayscale, flattened

Trained multi-class logistic regression to detect on our 36 classes

Yielded 16% test accuracy (random guessing is ~2.8%)

Model lacks spatial correlation and loses information through flattening, stronger baseline required

Baseline Model 2: From Scratch CNN

Created simple CNN, preserves spatial correlation

- ***Conv 1***: 3×3 Conv (3 → 32), BatchNorm, ReLU, 2×2 MaxPool
- ***Conv 2:*** 3×3 Conv (32 → 64), BatchNorm, ReLU, 2×2 MaxPool
- ***Conv 3:*** 3×3 Conv (64 → 128), BatchNorm, ReLU, Adaptive Avg Pool (1×1)
- ***Classifier:*** Flatten, Dropout (p = 0.3), FC Layer

Yielded 33% test accuracy

Model lacks feature extraction abilities of SOTA model with pretrained weights

Baseline Model 3: Fine-tuning ResNet-18

ResNet-18 pretrained on ImageNet

Final FC replaced: 512 → 36

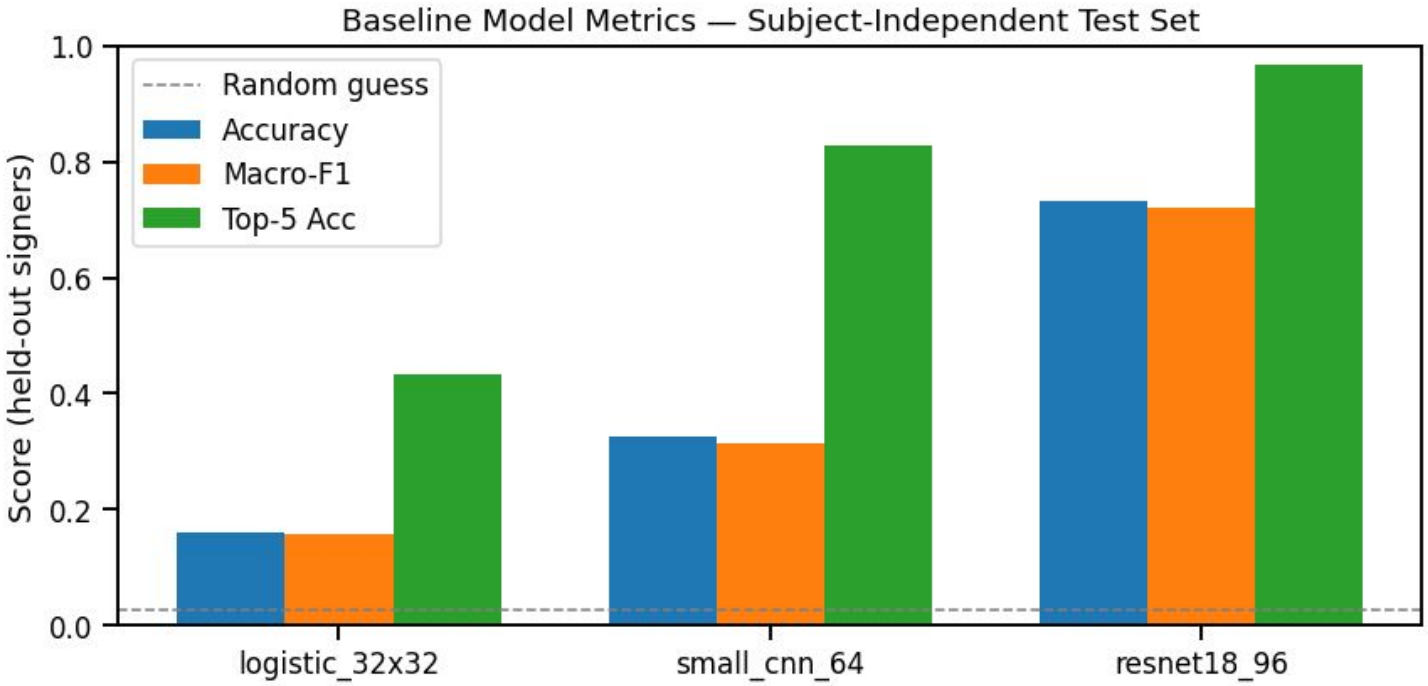
All layers fine-tuned, Adam 1e-3, 5 epochs, 96×96 RGB

~11M params total

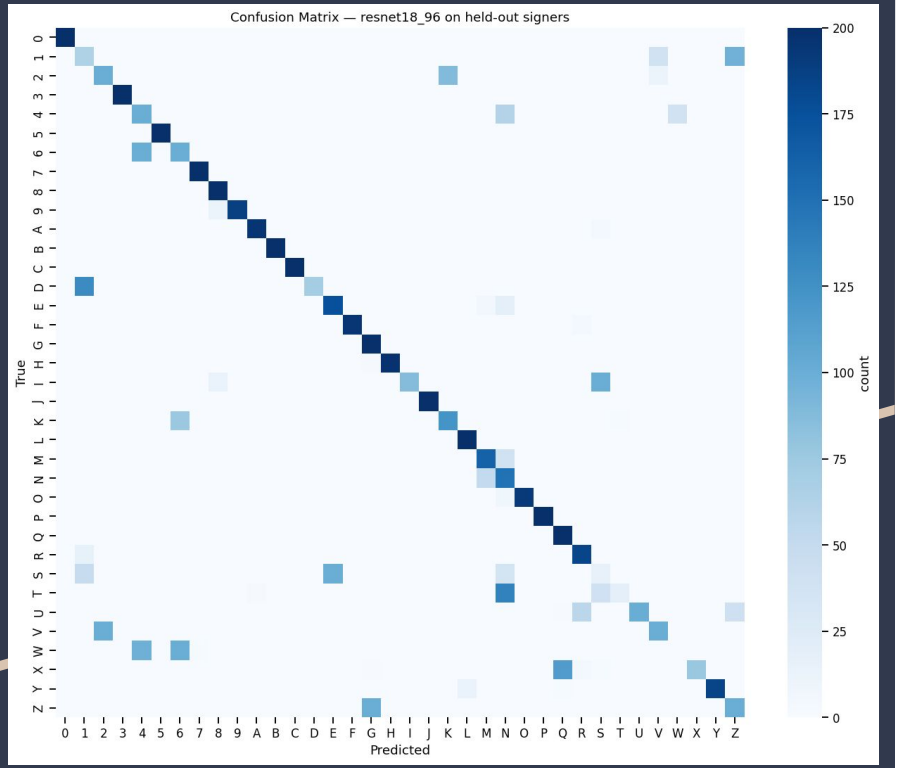
Yielded 73.3% test accuracy

Takeaway: ImageNet edge/contour features transfer well

Results


Baseline Model Metrics — Subject-Independent Test Set
Random guess
Accuracy
Macro-F1
Top-5 Acc
Score (held-out signers)
1.0
0.8
0.6
0.4
0.2
0.0
logistic_32x32
small_cnn_64
resnet18_96

Analysis

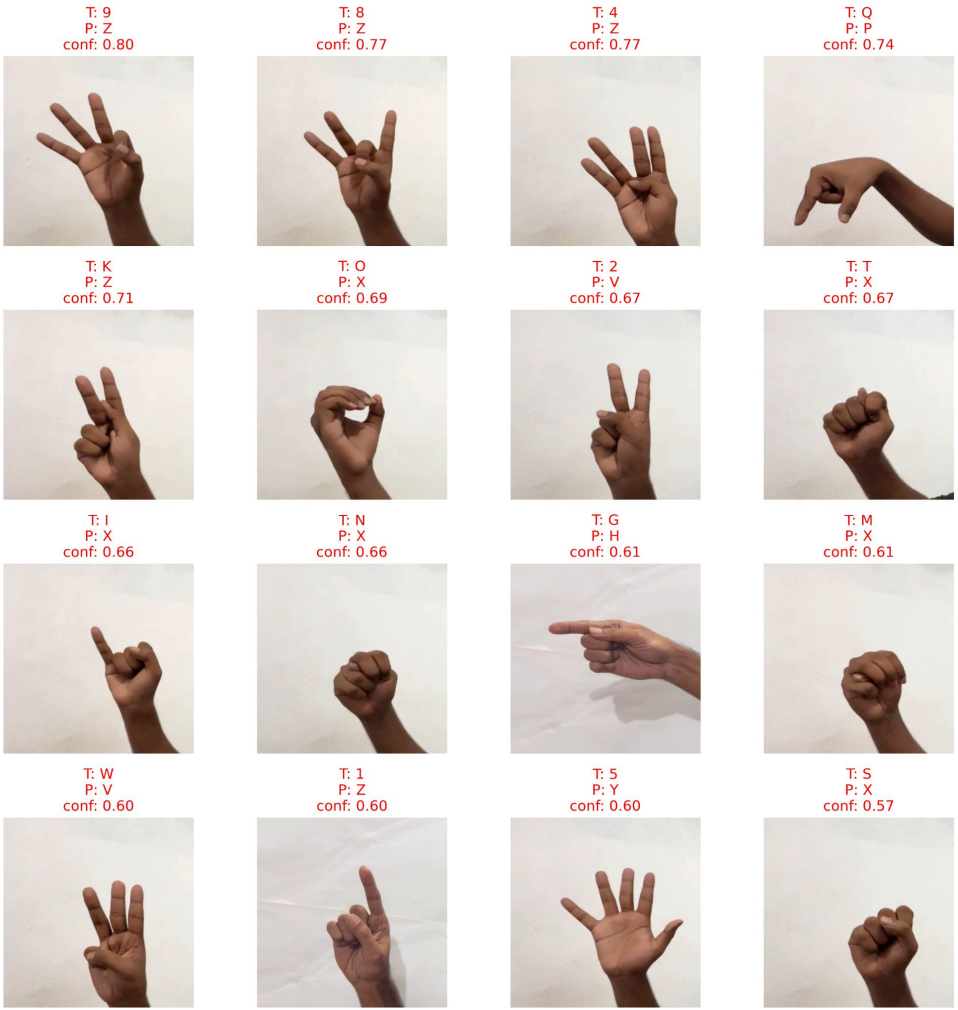


-The diagonal is quite strong, meaning the model is overall pretty good

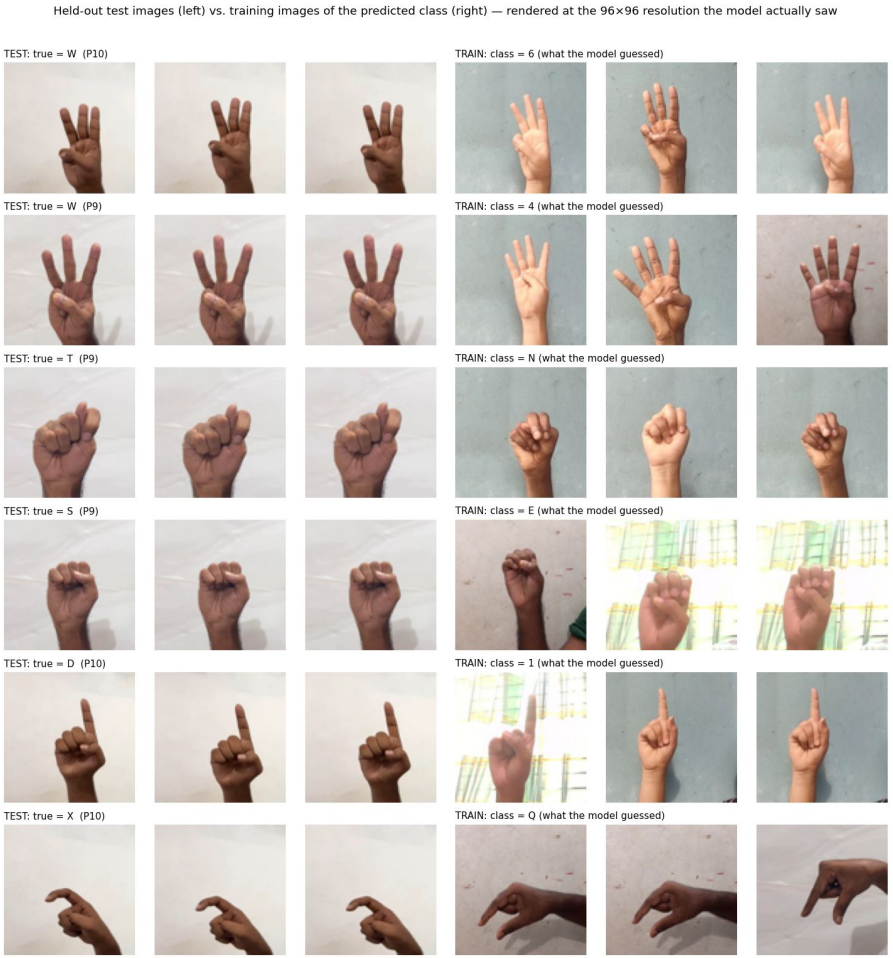
-But the errors are highly clustered on specific letters, which are generally have the same hand silhouette to other another from some angles but different finger placement.

-For instance 1/D is a big issue because they can look the same from some angles. Hand silhouette is the same, but finger placement is different

Most Confident Misclassifications



Looking into Misclassifications



Final Model Pipeline

1. Raw images
2. Subject-aware split
3. Preprocessing
4. Fine-tuning backbone
5. Evaluation

Future Steps

- Try more transfer learning models (ResNet50) + augment fine-tuning data, use a higher image resolution. (Hugh/Ayush)
- Fine-tune a Vision Transformer (ViT) to compare cross-subject generalization against ResNet18 (Hugh/Ayush)
- Experiment with much heavier augmentation (aggresive crops, aggressive color jitter) to improve robustness. (Zachary)
    - Potentially can augment images with Hand-landmark features (MediaPipe 21-keypoint)
- Meet in person to create out of sample test set with other subjects/backgrounds. We will try to meet on a sunny day for example. (All together)

Questions

-Is the current out-of-sample plan (self-collected images) sufficient, or should we also look at existing external ASL datasets for a more controlled comparison?

-If self collected images are sufficient do you have any ideas about ways we could “trick” our model is really push it to its brink